

Evaluating Automated System Interventions Against Email Harassment

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ABSTRACT

Email and other forms of electronic communication are becoming increasingly more essential to our everyday lives. However, with this growth comes the paralleled increased risk of email harassment, exacerbated by the current lack of platform support for managing these harmful messages. This paper explores different interfaces for the automated detection and management of email harassment using artificial intelligence in order to investigate what degree of platform intervention email users prefer when navigating their email platform. Through conducting a study involving three different email platform prototypes based on the Gmail platform, we evaluate how varying levels of platform intervention affect users' perceived sense of safety, agency, and trust with their email platform. Our primary findings suggest that users generally benefited from each of the system intervention strategies and desired higher intervention features when combating email harassment, as well as ways of managing this intervention based on their unique preferences.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Collaborative and social computing**; • **Social and professional topics** → **Computing / technology policy**; • **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

Online harassment, classification, labeling, support, automation, email

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1 INTRODUCTION

Individuals who rely on email as a central means of communication want to navigate their email platforms without being at

risk of harassment, i.e. messages that threaten their mental, emotional, or physical well-being. In 2017, around 41% of Americans had personally experienced online harassment – which included offensive name-calling, purposeful embarrassment, physical threats, and stalking – and among those victims, 10% had experienced their most recent incident of harassment through email [2]. Popular email platforms such as Google's Gmail contain some amount of system intervention in order to improve user experience, such as spam detection and the ability to create custom email filters. However, there is currently no automated way to robustly detect or flag harassment messages on these platforms before the user views them in their inbox, thereby placing the burden of managing harmful content onto the user.

In this study, we focus on determining what degree of intervention users prefer when interacting with AI systems that can detect harassment, and how these interventions affect users' sense of safety, agency, and trust with the system. We measure the degree of intervention of a given strategy based on the degree of separation between the user and the harassment email established by the system's actions. The three primary intervention strategies we examine are in-email warnings (low intervention), email tagging in inbox (medium intervention), and email filtering (high intervention). These strategies are based on existing interventions that Gmail currently employs for other types of undesirable emails such as spam. To evaluate these intervention strategies, we separated participants into three experiment groups – one for each strategy – and asked them to manage simulated harassment emails on the current Gmail platform and on a modified prototype version of the Gmail platform. After interacting with the control and prototype platforms, participants evaluated the intervention strategy by responding to a number of prompts regarding their sense of safety, agency, and trust with the system. Participants were also asked to elaborate on their experiences in a discussion-based portion of the study.

This paper primarily contributes insights to the HCI community regarding user perspectives on the automated management of email harassment that can guide the future development of more supportive and robust email platforms. The discussion of our findings highlight that generally (1) participants' sense of safety and agency benefited from each of the system intervention strategies, (2) user experiences with these strategies were highly dependent on the perceived accuracy of the detection system, (3) participants leaned towards higher intervention features, (4) participants sought further resources and protections from their email platform, (5) participants desired the ability to manage how their platform intervened against email harassment, and (6) any effective intervention strategy requires some degree of flexibility.

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2 RELATED WORK

There has been some amount of research regarding individuals' experiences with managing online harassment, as well as research conducted on how AI systems can effectively detect abusive online language through machine learning. However, these two areas of research have yet to be combined, and the question of how users would perceive and interact with an AI system that could detect harassment on email platforms has yet to be considered.

2.1 Experiences with Online Harassment

A significant amount of prior research, both within and outside of the HCI field, has focused on individual experiences with online harassment. Some of the most recent research within the U.S. highlight that online harassment is a growing issue, particularly for young people, women, and people of color [2]. Although most digital harassment occurs through social media platforms rather than through email, our research focuses on email given its intimate function and given the lack of interventions currently available on popular email platforms [2]. Outside of the U.S., online harassment is also a prevalent issue that varies in nature based on cultural and political influences. For instance, research interviews with women from South Asian countries illustrate the prevalence of cyberstalking, impersonation, and personal leakages within their communities, which have ultimately pressured these women into removing themselves from public online spaces [11]. Other research includes interviews with women from the Philippines who are legally unable to file charges against their online abusers, interviews with Colombian women who received repeated email threats from paramilitary groups due to ongoing regional conflict, and interviews with Pakistani women who are often victims of hate speech online revolving around blasphemy and national identity [4]. Though our research primarily focuses on harassment within the U.S., this work provides a broader understanding as to how experiences of online harassment are heavily shaped by cultural influences.

Some of this research has also examined potential technical strategies that platforms could employ for managing online harassment, such as offering custom filtering, providing alternatives to deactivation, and taking more swift and effective action against harassers [14]. While this prior research has not focused on AI system interventions, these explorations aid in the design and execution of our research.

2.2 Detecting Abusive Language with AI

Over the past few years, numerous methods for identifying abusive language using artificial intelligence have been investigated. A substantial component of the detection process is the collection and categorization of online harassment data since determined categories prove to vary across studies. Categorizations may be based on a predefined list of offensive words, on the severity of the written harassment, on whether the text negatively targets an aspect of identity (ex: race, gender), or on a number of different measures combined [3, 5, 12]. Another primary facet of this research is the creation of machine learning models that can effectively detect abusive language, which involves the examination of various ML algorithms and architectures [3, 7, 12]. One tool that is currently

in public use for detecting abusive language online is Jigsaw's Perspective, an API that uses machine learning to indicate how likely a given piece of text is to be perceived as "toxic" across multiple definitions of toxicity (based on a database of human-evaluated internet comments) [10]. While our research predominantly focuses on the design of the user interface for an AI system that detects harassment emails, the aforementioned research and current tools available suggest that such a system would be possible to implement, as well as offer some insight into what kinds of harassment this system could recognize.

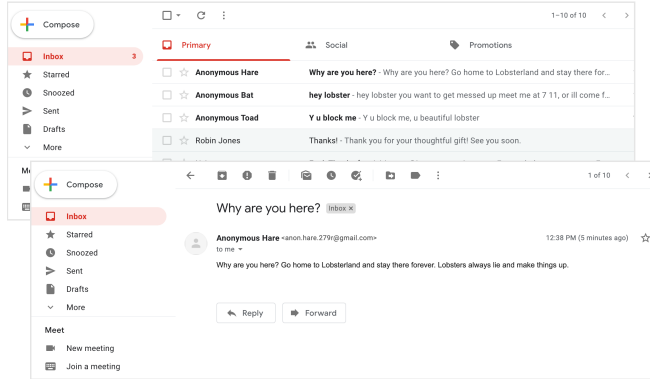
2.3 Current Approaches for Automated Email Intervention

Popular email platforms like Gmail already employ intervention features like warning labels and filtering for undesirable emails such as spam. However, these interventions do not currently extend to harassment emails, though users may manually create filter rules based on fields such as selected keywords, sender information, and email size. There has also been some research conducted relating to automated email intervention against phishing, which reveals the effectiveness of reporting, blocking, warning, and embedded education as intervention features [13]. Specifically, authoritative warning messages that employ clear and concrete language prove to be the most effective when managing phishing and malware emails [9]. While these methods provide useful insight into how undesirable emails can be managed, designing autonomous system interventions against email harassment presents a number of unique challenges that differ significantly from those presented by malware and phishing, particularly given the often personal and sensitive nature of harassment.

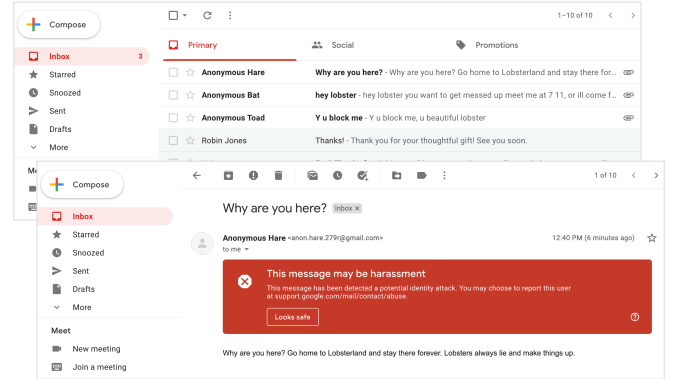
2.4 Current Approaches for Email Harassment Management

There are currently a few services that aim to protect users from online harassment within their inboxes, none of which employ automated detection and flagging strategies. For instance, HeartMob is a web platform that organizes a community of volunteers who provide support to individuals experiencing online harassment, along with resources relating to online privacy and protection [1]. HeartMob equips victims of harassment with a valuable support network, which predominantly focuses on the aftermath of receiving a harassment message. Our study aims to address a separate user need, namely the ability to intercept email harassment before it affects the individual. Another approach is Squadbox, a friend-sourced moderation service that allows email users to set up filters within their inbox to detect suspicious emails, forward detected emails to a selected group of friends or moderators, and then rely on those individuals to determine whether the intercepted email should be returned to the user's own inbox [8]. Squadbox is effective at providing community-based support to online harassment victims but still requires human intervention to manage a user's harassment emails. Our study examines the design of automated platforms that retain the privacy of the user's email inbox (which may be desirable in many cases where harassment involves sensitive information) and that reduce the logistical and emotional burden of moderating abusive language in its entirety.

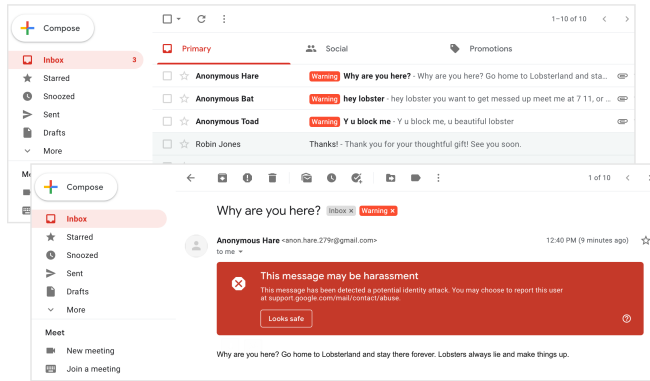
Control (1)



Prototype A-WARN (2)



Prototype B-TAG (3)



Prototype C-FILTER (4)

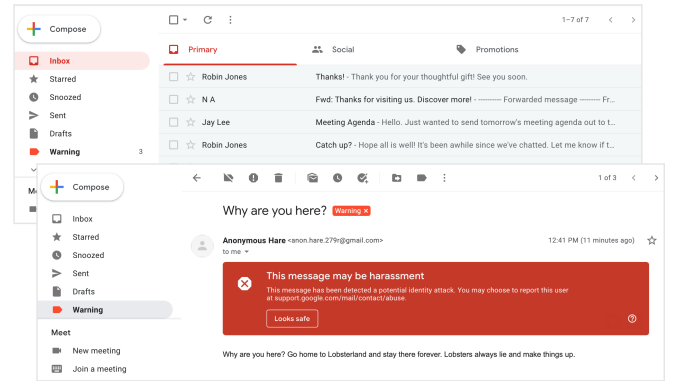


Figure 1: Designs of the four email platforms, including Control (1) which implements no system interventions, Prototype A-WARN (2) which implements in-email warnings, Prototype B-TAG (3) which implements email tagging, and Prototype C-FILTER (4) which implements email filtering.

3 PROTOTYPE DESIGN

To assess how varying intervention strategies would affect the users' sense of safety, agency, and trust with their email platform, three prototype platforms were designed to simulate the actions of a hypothetical harassment detection AI system, as shown in Fig. 1. For a basis of comparison, participants interacted with a **Control** email platform which implements no interventions against harassment (i.e. the current Gmail platform). The following prototypes do not involve a fully-implemented AI system. Instead, these prototypes rely on the manual configuration of existing Gmail features, such as custom email tags and filters, in order to simulate the behaviors of the hypothetical AI system.

- **Prototype A-WARN: In-Email Warning (low intervention).** Prototype A-WARN implements a red warning label at the top of every email detected as harassment by the AI system. This warning label is modeled after a similar label that Gmail includes for emails detected as spam and authoritatively presents a means of action in clear and concrete

language as inspired by pre-existing email intervention design research [9]. The warning label includes information as to why the email was detected as harassment and further actions the user could take to report the sender. The user also has the option to indicate that a detected email should not be categorized as harassment by marking the email as "safe." This feature was simulated through the use of an inline image within the harassment email.

- **Prototype B-TAG: Email Tagging in Inbox (medium intervention).** Prototype B-TAG implements red warning labels next to emails detected as harassment within the inbox, before the user opens the email. Prototype B-TAG also includes the same red warning label from Prototype A-WARN within the opened email.
- **Prototype C-FILTER: Email Filtering (high intervention).** Prototype C-FILTER implements a separate folder where emails detected as harassment are filtered into, rather than remaining in the inbox. This folder is located in the left sidebar of the Gmail interface, along with other email folders

such as spam. Prototype C-FILTER also includes the in-email warning from Prototype A-WARN and the red warning labels from Prototype B-TAG.

3.1 Simulating Harassment

Due to the sensitive nature of online harassment, we artificially defined offensive content with neutral vocabulary. For the purposes of this study, participants were instructed to consider the word “lobster” to be offensive to their identity and values. This specific term was selected because it does not have any offensive associations in American culture, and we opted for an English word to avoid any confounding effects of using a foreign language that could have certain cultural associations depending on the participants’ backgrounds. Additionally, participants were instructed and encouraged to interrupt at any point if they felt uncomfortable with the language used in the study.

For the user study, three harassment emails were designed and added to the inboxes on each email platform (the three prototype platforms and the control platform), all from separate anonymous email accounts. The contents of each email were based on sample harassment data used by the Perspective API [10]. The three types of harassment emails shown – identity attack, threat, and flirtation – were selected to encompass a broad range of abusive language given the distinct nature of each category and their frequency in the Perspective API sample harassment database. For instance, the simulated harassment email reflecting an identity attack read: “Why are you here? Go home to Lobsterland and stay there forever. Lobsters always lie and make things up.” Only three types of harassment were simulated in order to avoid overwhelming the user. Additionally, seven non-harassment emails were included and marked as read on each email platform, in order to simulate a more realistic email inbox.

4 STUDY DESIGN

4.1 Participants

We conducted nineteen video call interviews in order to assess users’ interactions and perspectives with each of the three prototype email platforms. We primarily recruited participants by directly emailing undergraduate college students who had active online presences. The remainder of our participants were recruited through personal networks. All of the participants were current or recently graduated students from U.S. colleges whose ages ranged from 18 to 22, and none of the participants received any compensation. Of the nineteen participants, six interacted with Prototype A-WARN, six interacted with Prototype B-TAG, and seven interacted with Prototype C-FILTER. Throughout our recruitment process, we remained mindful that young people, women, and people of color are disproportionately affected by online harassment [2], and we ensured we gathered insights from these particular demographic groups. Each of the three experiment groups included at least four participants who identified as female, at least three participants who identified as different ethnicities, and at least two participants who had prior experience with online harassment.

4.2 Structure

This mixed-method user study consisted of six stages. First, participants completed an anonymized form that collected information about their demographics and online habits. Second, participants interacted with the control platform by logging into a pre-configured Gmail account containing the three simulated harassment emails, navigating the inbox as if it were their own, and describing their thought process aloud for each of the actions they executed. Third, participants evaluated the control platform by anonymously indicating the extent to which they agreed or disagreed with a number of statements relating to safety, agency, and trust with the platform (using a 7-point Likert scale) and were invited to explain their reasoning out loud. Fourth, participants followed the same procedure with their designated prototype platform by logging into another pre-configured Gmail account and viewing the same inbox as in the control scenario, except where the harassment emails were indicated based on the implemented intervention strategy. Fifth, participants again anonymously evaluated a number of statements relating to safety, agency, and trust with their assigned prototype platform and were invited to explain their reasoning out loud. Lastly, in a discussion-based debrief, participants were asked about their general experiences using both platforms, as well as what platform features they believed would be most useful to them when managing harassment emails. All of the aforementioned data was collected for educational purposes through a research course project and has not been reviewed by the Harvard Institutional Review Board.

5 FINDINGS AND DISCUSSION

In this study, we define safety to mean how protected from harm participants felt while navigating their email platform, we define agency to mean how much control participants felt over the management of their emails, and we define trust to mean how much participants believed their email platform to be effective, dependable, and well-intentioned.

5.1 User Safety and Agency Increase with System Intervention

Users’ sense of safety and agency generally increased with each of the implemented intervention strategies. When compared to the Control platform, a greater number of participants agreed with the following statements after interacting with their prototype email platform: “I feel comfortable continuing to use my email platform” (68.4% v. 78.9%) and “I have sufficient control over how I manage harassment messages” (47.4% v. 63.2%). For those who experienced an increased sense of safety or agency, the system intervention on their prototype platform was viewed as meaningful and empowering, thereby functioning in the users’ best interest. One participant who interacted with Prototype B-TAG stated, “*having the interface acknowledge that [the email] is harassment... makes it feel safer*” and appreciated the platform being “*more direct about taking action*.” Another participant who interacted with Prototype A-WARN expressed feeling as if “*[their] experience [of harassment] mattered*” when navigating the prototype platform. Overall, these results suggest that these three intervention strategies would greatly benefit many users’ sense of safety and agency while navigating their email platform.

5.2 System Accuracy Significantly Impacts User Agency and Trust

Though system intervention was illustrated to benefit many users' sense of agency and trust while navigating their email platform, many participants noted that these feelings would be dependent on the accuracy of the system. Multiple participants (36.8%) remarked that if the system were to misclassify emails often – either failing to detect harassment emails or detecting non-harassment emails as harassment – their sense of control over how their emails are managed would feel infringed on, as well as their overall trust in their email platform's ability to manage emails effectively. One participant who interacted with Prototype C-FILTER stated that *"my sense of trust is more tied to the ability of the platform to actually filter out harassment emails."* Another participant from the Prototype C-FILTER experiment group commented that their sense of agency *"depends on the accuracy and how [the platform] classifies things because [it] has a potential to be more hassle."* Participant concerns over system accuracy are certainly well-founded, considering that even industry-leading harassment detection technologies like Perspective API can still be manipulated or abused [6].

5.3 Users Lean Towards High Intervention Platform Features

Across all of the prototype platforms, participants generally wanted higher system intervention features to combat email harassment. Half of the participants who interacted with Prototype A-WARN desired further indication from the system that an email was detected as harassment, preferably before the participant had to open their email. One participant commented that *"when looking at the inbox, [they] should not have to open [the email] if [they] don't want to."* One third of the participants who interacted with Prototype A-WARN and Prototype B-TAG also wanted further system intervention, notably in the form of some kind of filtering. Lastly, a majority of participants who interacted with Prototype C-FILTER (71.4%) desired additional intervention features such as the ability to block or report email addresses without having to open the detected email. These qualitative user sentiments indicate that Prototype C-FILTER employed the most effective intervention strategy out of all of the prototype options, however it is evident that a more effective platform would implement even higher intervention features.

5.4 Users Desire More Resources And Protections

Many participants indicated that they would want their email platform to provide more resources in the case where the user received email harassment. While all of our prototype platforms provided a reporting link from the in-email warning, multiple participants (21.1%) desired additional information such as more details behind the reporting process, how users could contact the local police for help if needed, and general resources relating to mental health support. One participant who interacted with Prototype B-TAG noted, *"the link for reporting wouldn't be enough for me... having another direct link to other points of action or resources would be helpful."* Existing solutions like HeartMob also illustrate the clear user need

for further logistical and emotional support when managing email harassment [1].

5.5 Users Want to Manage How Their Platform Intervenes

Many participants desired further control over how the system detected and managed harassment emails, with a majority of participants (78.9%) agreeing with the statement: "It is important for me to manage how my email platform detects and intercepts harassment emails." A number of feature ideas proposed by participants revolved around customizing how email harassment should be managed, such as filtering certain emails by selected keywords or choosing whether certain types of harassment emails should be managed differently (e.g. if the user wanted to be notified of threats but would rather filter out identity attacks). Some participants (15.8%) also indicated a desire to directly train the AI system to better detect harassment emails by correcting mistakes or validating accurate classifications.

5.6 Any System Intervention Strategy Requires Flexibility Given Range of User Preferences

Participants held a broad range of opinions when evaluating the following statements: "It is important for me to manage harassment emails directly" and "It is important for my email platform to manage harassment emails on my behalf." Participants also demonstrated a broad range of email habits while interacting with the control and prototype platforms; whether participants chose to leave emails unread, to delete emails, to archive emails, or to report emails as spam varied widely based on their individualized email management preferences. One participant noted, *"I would rather not see [the harassment email]. I know some people would rather read it, but I would rather not."* This range indicates how any system intervention designed to combat email harassment requires some degree of flexibility in order to avoid imposing on users' pre-existing routines and on their perceived relationship to their email platform.

6 LIMITATIONS AND FUTURE WORK

Our user study presented a number of limitations, as well as multiple areas for future development. Because all participants first interacted with the control platform, participants' interactions with their prototype platform were shaped by having prior experience with simulated harassment. In other words, we did not obtain evaluations of our prototype platforms from any users with zero experience with receiving a (simulated) harassment email. The simulated harassment emails were also the same across the control and prototype platforms, which may have influenced how users interacted with the prototype platforms, such as their inclination to open the emails again. Additionally, the simulated harassment emails were all authored by anonymous senders. Because multiple participants noted that their management of the harassment email would depend on whether or not they knew the harasser, future iterations of this research could evaluate automated system interventions against a greater range of online harassment sources, including simulated harassment authored by individuals familiar to the participants. Another area of future work would be investigating users'

responses to system inaccuracies with email harassment detection, given that multiple participants reported that their willingness to use an AI system for email harassment management would vary according to the system's accuracy. Lastly, because most participants indicated an interest in higher levels of system intervention, it would be valuable to explore the upper limits of acceptable system intervention in order to better understand the extent to which users would be willing to rely on AI systems for email harassment management.

7 CONCLUSION

This paper examines participants' perceptions of three different intervention strategies for the automated management of harassment emails. We conducted a user study with nineteen participants to identify how varying levels of system intervention affect users' sense of safety, agency, and trust with their email platform. Our results indicate that these system intervention strategies generally improved participants' experiences on their email platform, with participants leaning towards higher intervention features and seeking ways to further manage how their platform intervened. Based on these insights, an email platform that assists users in the management of harassment emails would implement high system intervention features that combines in-email warning messages, email tagging, and email filtering. This platform would also provide users with sufficient resources in the event that they receive harassment and allow users to customize different management features to suit their pre-existing email routine.

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